

Introduction

The global geography of scientific research has changed a lot in the past century. The international movement of scientists is widely seen as an important force behind knowledge exchange, innovation, and the strengthening of research competitiveness. Understanding where elite scientists choose to conduct their research can reveal structural advantages in the global scientific system. Nobel Prize laureates are a useful group for studying these changes. Because they are very important scientists, looking at where they were born and where they did their main research can show us mobility of these elite scientific talent.

Model Used:

- Temporal trend analysis model examines how the scale and direction of scientific mobility have changed across different historical periods.
- Social Network Analysis models can analyse which countries act as major “senders” or “receivers” of scientific talent. Also using Louvain community detection to identify regional clusters in the global talent mobility network.

Research Question

Main Research Question (MRQ):

How has the global geography of Nobel Prize-winning research changed from 1901 to 2024, and what patterns can we see in how laureates move to different countries to conduct their research?

RQ1:

Which countries are the primary 'Brain Gain' nations, which are the primary 'Brain Loss' nations, and what are the major geographical patterns and trends in this talent flow

RQ2:

How has internationalization evolved? Have dominant destinations or mobility routes shifted across historical periods?

Data Collection

Data Source:

We obtained data from Wikidata via SPARQL queries, covering all Nobel Prize categories awarded 1901-2024.

For each laureate, we collected:

- Name and award year
- Birth country
- Employer institutions with start and end dates
- Institution countries

Data Cleaning

To determine the main research country for each laureate, which is not directly provided in Wikidata, we use their employer institutions, along with the start and end dates and the countries of these institutions. Below is the process:

Algorithm Priority Logic:

1. Select institutions whose employment overlaps the award year
2. Among overlapping institutions, choose the longest-duration employer
3. If no overlap, select the longest-duration institution
4. If all employer institutions have no temporal data, use most frequent country that appear

To ensure consistency in the analysis, we mapped all historical political entities to their corresponding modern country borders. Without this mapping, the same geographic region would be mistakenly counted as multiple instances of international mobility, distorting the true talent mobility patterns. For example:

- "Kingdom of Italy": "Italy"
- "Nazi Germany": "Germany"
- "Soviet Union": "Russia"

Final Dataset:

- Total records : 847
- Total unique laureates: 842
- Time span: 1901–2024
- Countries represented after mapping: 75

Libraries

- SPARQLWrapper for querying Wikidata
- Pandas for data manipulation
- NumPy for numerical computations
- NetworkX for network construction
- Matplotlib for visualization
- Seaborn for visualization

Model/Method & Implementation

Social Network Analysis

For Research Question 1, I choose the Social Network Analysis models because it helps describe how Nobel laureates move across countries during their scientific careers. It shows the whole network of connections between countries. The model is constructed by node,

directed edges and edge weight where:

- Node: countries
- Directed edges: mobility route
- Edge weights : number of laureates on each route

To implement this model, we first extract all records where the birth country is not equal to the main research country. We then use group-by to count the number of laureates along each country-to-country route, and afterwards use NetworkX to create a directed graph (DiGraph). And below is our key metrics to characterize countries' roles in the global talent system:

1. In-Strength: The number of foreign-born Nobel laureates who moved to a country and conducted their prize-winning research there.
2. Out-Strength: The number of Nobel laureates born in a country who carried out their award-winning research somewhere else.
3. Net Cross-Border Flow(In-Strength - out-strength) : It shows whether a country is a net importer or exporter of scientific talent.
4. Total Talent Stock : All Nobel laureates who ended up researching in the country, including:those born there (and stayed), and those who arrived from abroad.

To further analyse the structure of global scientific mobility, I also applied community detection using the Louvain modularity optimisation method. This step helps identify groups of countries that are more closely connected through Nobel laureate mobility. To implement it , I switch the graph to undirected networks since this is the standard approach.After that we will get the modularity value and the community.

Temporal trend analysis model

In Social Network Analysis model, we will know about the spatial patterns of talent flows, but cannot explain how these patterns change over time. Therefore, using temporal trend analysis model is easier to capture the long-term shifts in global research mobility and explain why some countries become brain-gain destinations while others experience brain-loss.

To implement the temporal model, we first identify the the top five countries with the highest total number of Nobel laureates. All remaining countries are grouped into a single Other category.This step is necessary because when we examine the geographical distribution across many decades, a long list of countries would produce a fragmented pattern that is difficult to interpret and hard to visualize using graph.

In addition to analysing mobility patterns by decade, I also divide the timeline into three major historical periods—1901–1945, 1946–1990, and 1991–2024. These periods roughly correspond to key phases in global scientific development:

- 1901–1945: the early formation of modern science and the dominance of Europe before World War II;
- 1946–1990: the postwar expansion of science, the rise of the United States, and Cold War geopolitical influences;
- 1991–2024: the era of globalisation, increased international mobility, and the emergence of new research centres.

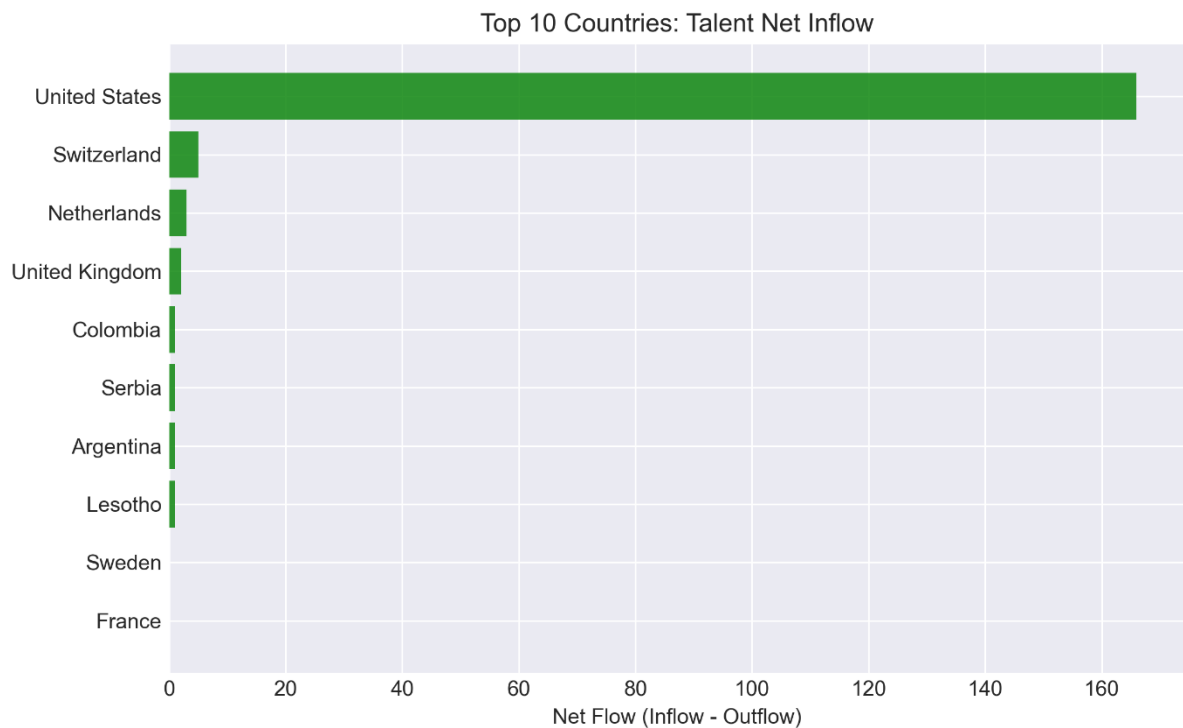
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Dataset is grouped by decade to compute mobility rates and the changing geographical distribution of Nobel Prize-winning research. Also, the three historical periods are then used to identify broader structural changes of geographical distribution of Nobel Prize-winning research.

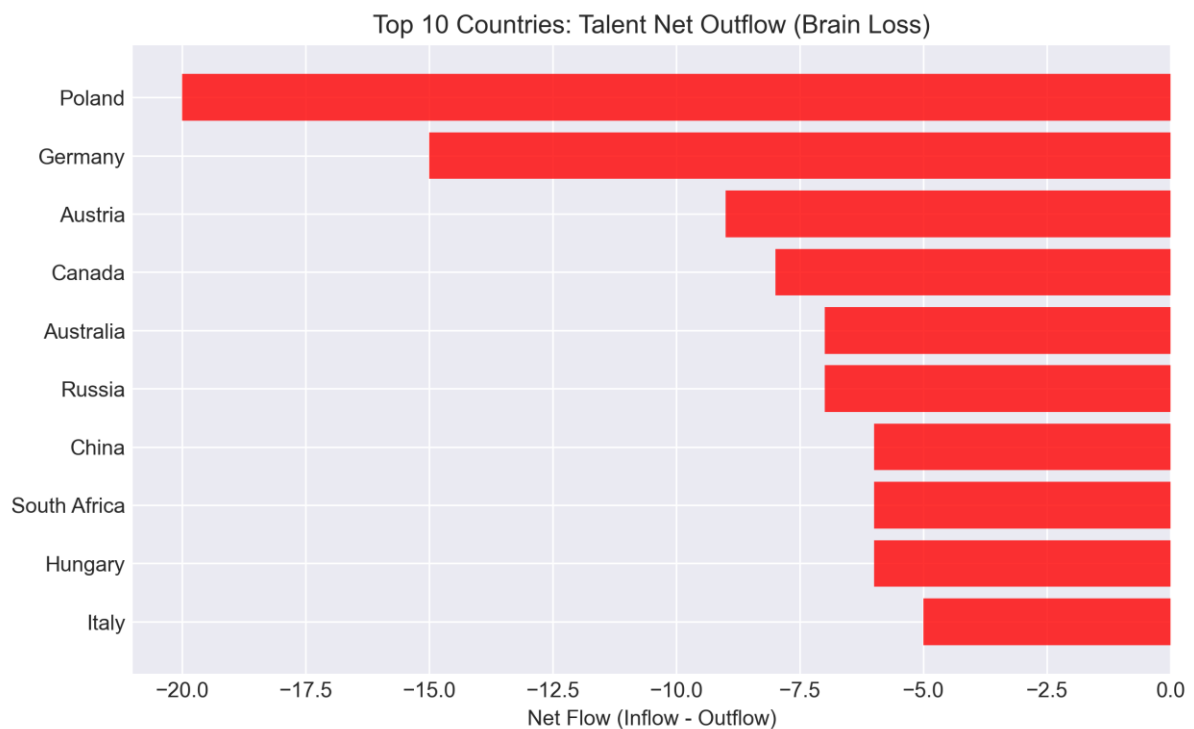
To deepen the analysis, I also include two additional indicators that reveal patterns not visible through simple share trends. The first is a concentration index calculated for each decade, which shows how geographically centralised Nobel Prize-winning research is. The concentration index was computed by summing the squared shares of each country within a decade. This helps identify whether scientific activity is dominated by a small group of countries or spread more evenly across the international system.

The second indicator is the volatility of each country group's research share over time. Volatility was calculated as the standard deviation of each country group's decade-by-decade share. This captures how stable or changeable a country's scientific position has been across different decades. Higher volatility indicates sharper rises or declines in influence, while lower volatility reflects a more consistent and steady contribution.

Results

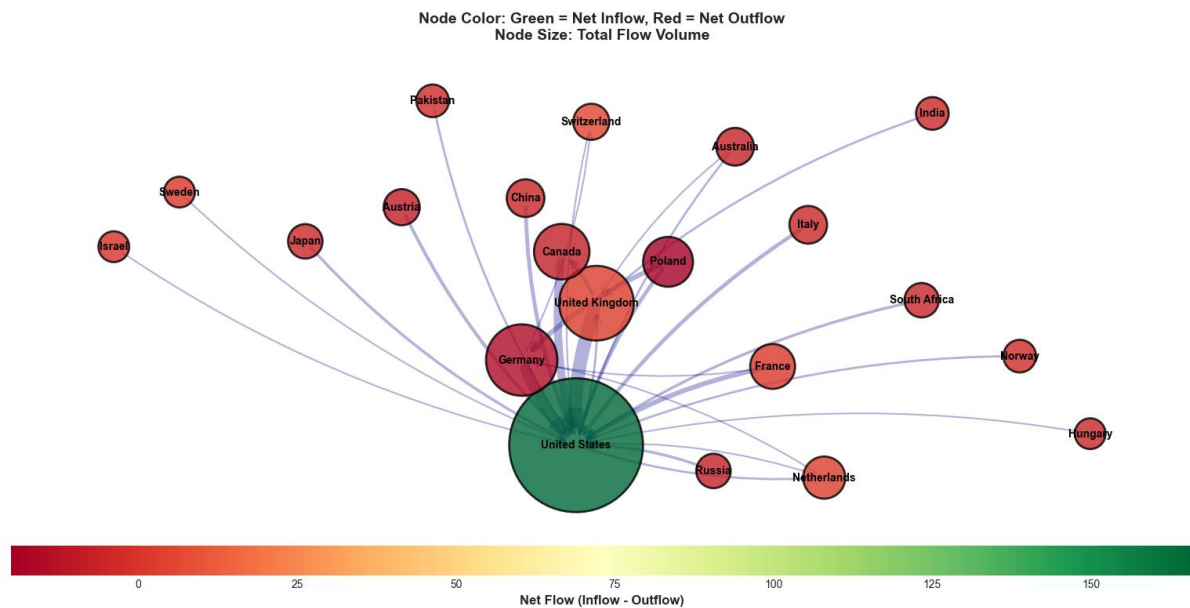


This chart illustrates net attractiveness at the national level. The United States net inflow far exceeds that of other countries, demonstrating its structural advantage as a global research hub. With more than 160 net inflows, United States stands as a dominant magnet that concentrates scientific talent.

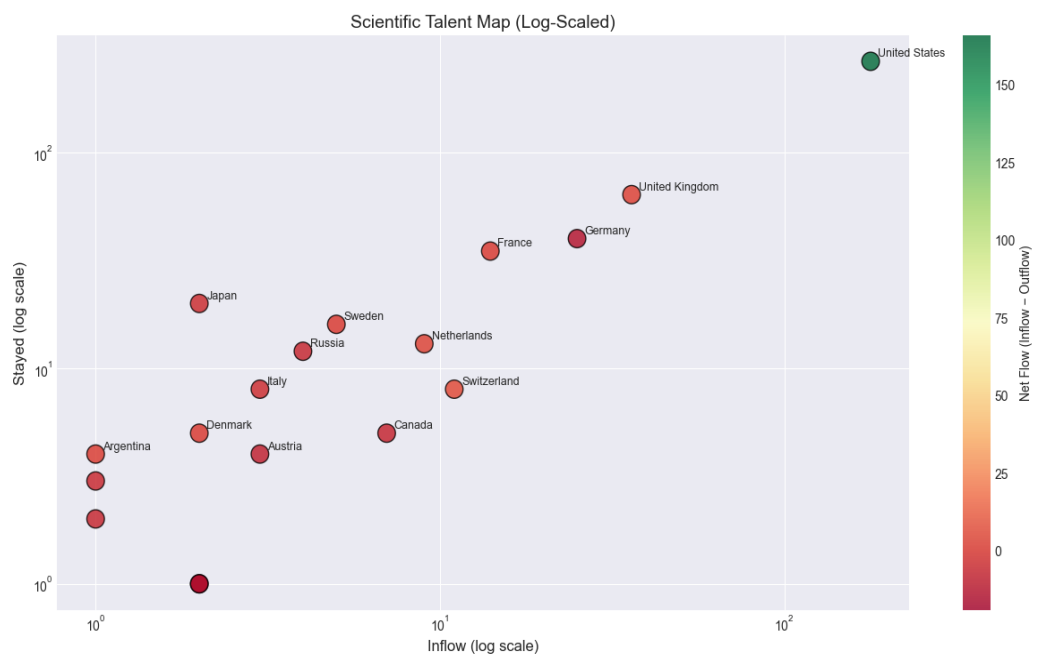


This chart highlights the top countries experiencing talent net outflow, often referred to as brain loss. Poland shows the largest net talent deficit, followed by Germany and Austria. Compared with the inflow chart above, the outflow distribution is far less extreme.

By contrast, losses in other countries are much more evenly distributed, and no single country experiences outflow on a comparable scale.



United States occupies a clear central position: almost all major mobility edges converge toward it, forming a strong core-periphery structure. The United States occupies the central core of the global scientific system, with almost all major flows directed toward it. Other countries form the surrounding periphery, connected to the core but not strongly connected to each other. This network structure visually demonstrates the asymmetry of global scientific mobility

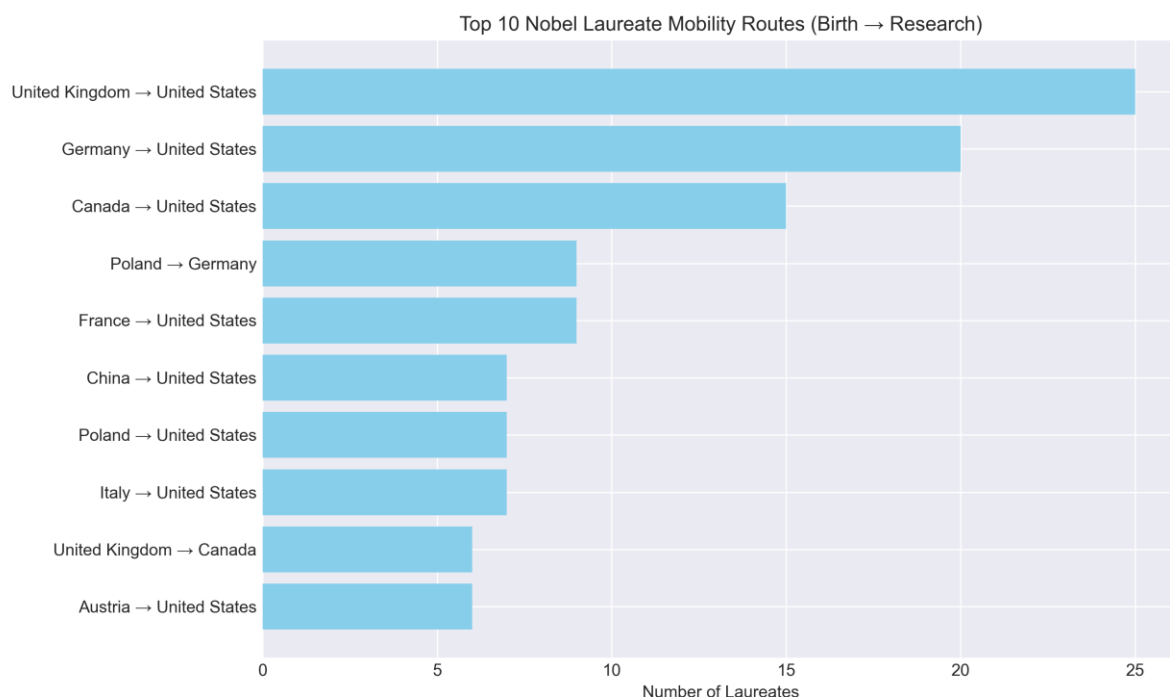


The log-scaled talent map highlights differences in how countries build their scientific strength.

The log-scaled talent map highlights a different type of country, such as Argentina and Japan. These countries have many laureates who stay in their home country and only a small number of foreign researchers coming in. This suggests that their Nobel achievements are mainly supported by strong and independent national research systems, rather than by attracting international scientists.

The log-scaled talent map also shows that the United Kingdom and Germany act as important secondary hubs. Both countries attract talent and also keep many home-grown researchers, which reflects their long scientific traditions.

Switzerland and Canada also emerge as notable mid-level hubs, but their role in the network is slightly different. Both countries attract more foreign researchers than they produce domestically, indicating that their scientific strength is partly built on international inflow



The figure shows the top ten Nobel laureate mobility routes from birth country to main research country. The United States appears as the dominant destination, receiving the largest flows from the United Kingdom, Germany, Canada, France, China, Poland, Italy, and Austria. Only one major route does not involve the United States as the destination: Poland → Germany.

=== Community Division Results (Louvain) ===:

Community 0 (Number of Nodes: 31):

['Austria', 'Bangladesh', 'Canada', 'Chile', 'China', 'Costa Rica', 'Czech Republic', 'Ghana', 'Ireland', 'Italy', 'Japan', 'Kenya', 'Lebanon', 'Lesotho', 'Lithuania', 'Myanmar', 'Nigeria',

'Norway', 'Pakistan', 'Palestine', 'Peru', 'Serbia', 'Slovenia', 'South Africa', 'Spain', 'Taiwan', 'Trinidad and Tobago', 'Tunisia', 'Turkey', 'United States', 'Venezuela']

Community 1 (Number of nodes: 10):

['Bosnia and Herzegovina', 'Colombia', 'Croatia', 'Germany', 'Indonesia', 'Latvia', 'Netherlands', 'Poland', 'Romania', 'Switzerland']

Community 2 (Number of nodes: 15):

['Australia', 'Azerbaijan', 'Brazil', 'Cyprus', 'Egypt', 'India', 'Liberia', 'New Zealand', 'Russia', 'Saint Lucia', 'South Korea', 'Tanzania', 'Timor-Leste', 'United Kingdom', 'Zimbabwe']

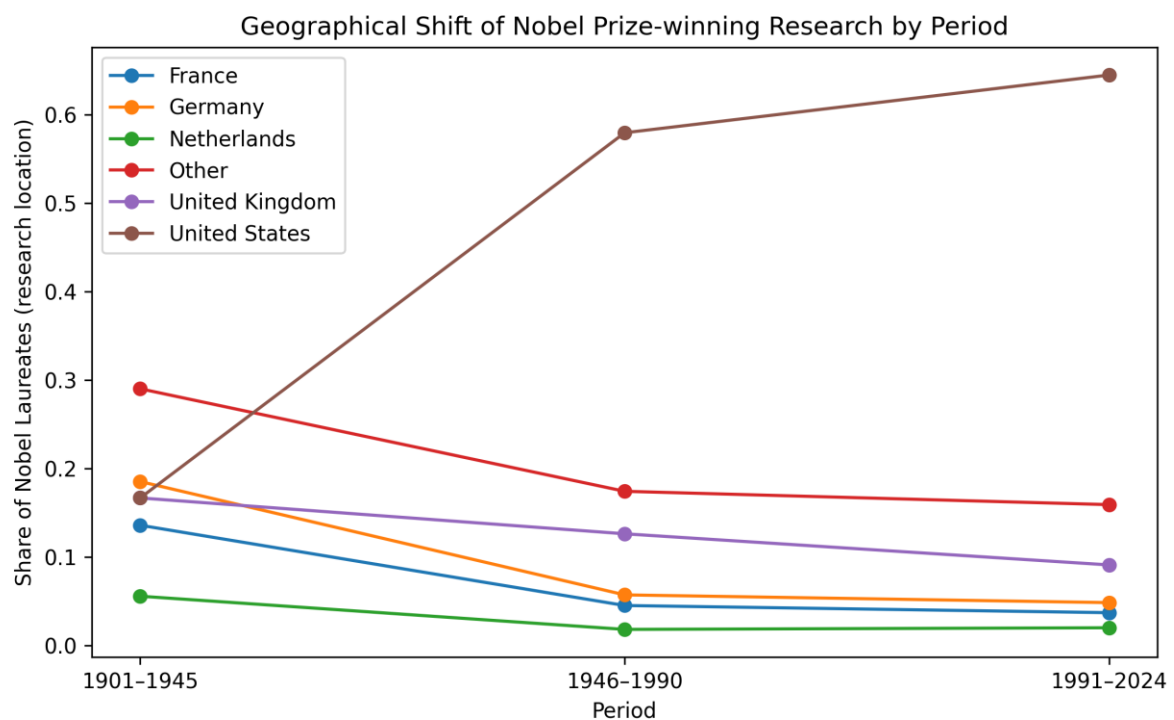
Community 3 (Number of nodes: 10):

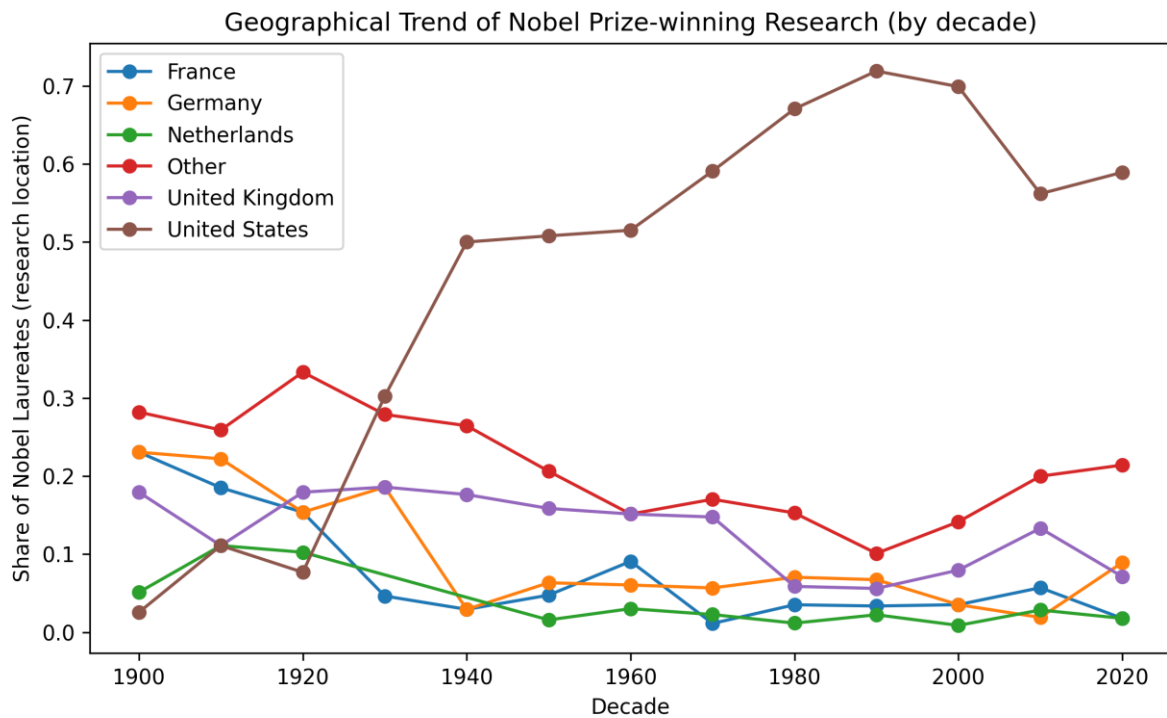
['Algeria', 'Argentina', 'Belgium', 'Denmark', 'Finland', 'France', 'Luxembourg', 'Mexico', 'Morocco', 'Sweden']

Community 4 (Number of nodes: 5):

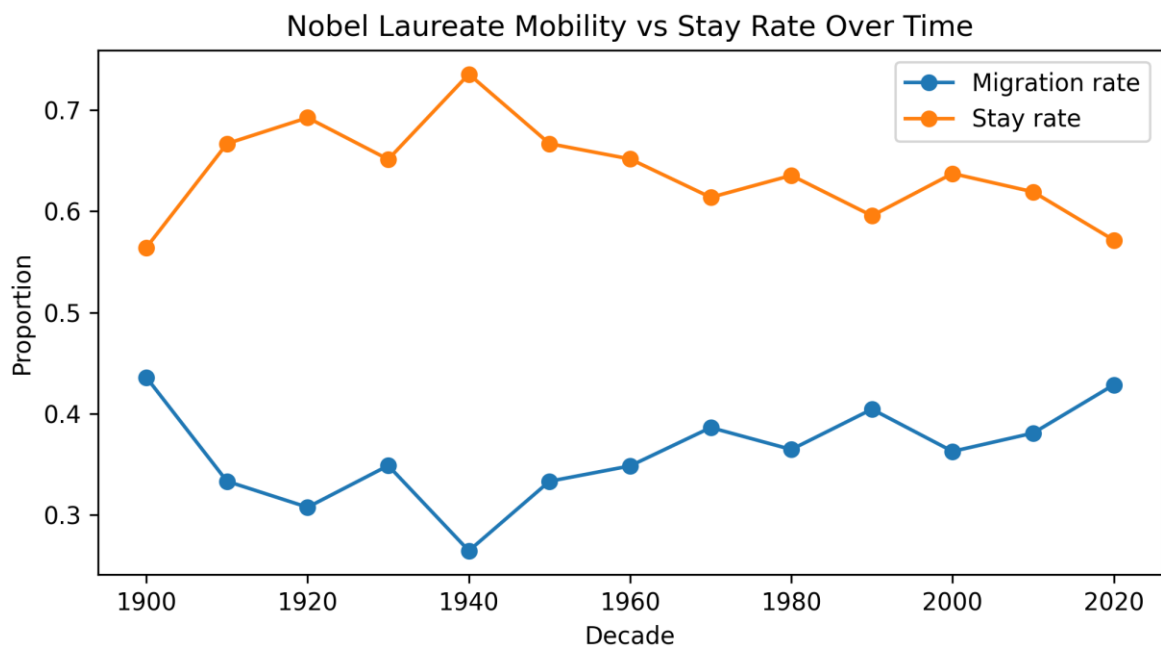
['Belarus', 'Hungary', 'Israel', 'Slovakia', 'Ukraine']

The Louvain community detection algorithm produced a modularity score of $Q = 0.198$.





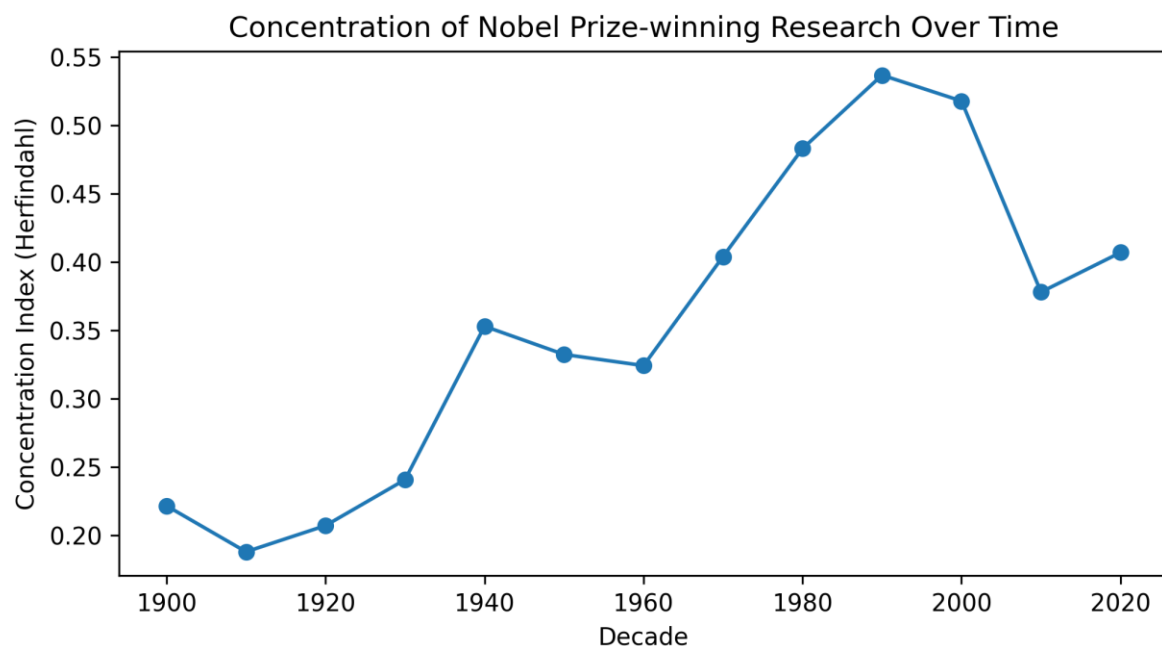
The geographical trend figure illustrates a major shift in where Nobel-winning research takes place. At the beginning of the century, European countries such as France, Germany, and the United Kingdom were the main centres of scientific activity. Over time, however, their share gradually decreased, while the United States rose sharply and became the dominant location for Nobel-level research. In the later decades, the U.S. consistently held a much larger share than any other country.



The mobility trend graph shows that international movement has always been a part of Nobel

laureates' careers, but its importance has changed over time. In the early decades of the 20th century, mobility was relatively high, then declined around the mid-century period, and later rose again. In recent decades, mobility has remained stable at a moderately high level, suggesting that cross-border research careers have become a normal feature of modern science.

Overall, the two graphs together reveal a long-term transformation in the global research landscape. Although mobility patterns changed over the years, the most striking development is the movement of scientific leadership from Europe to the United States. By the second half of the century, the U.S. had become the central hub of Nobel Prize research, and it has maintained that position ever since.



This graph shows how the geographic distribution of scientific leadership has evolved over time. The concentration remains relatively low in the early twentieth century, but rises sharply from the 1940s onward, reaching its highest levels in the 1980s and 1990s since the United States dominates Nobel-level output at that period. After 2000, the index drops, suggesting a period of greater globalization.

Volatility of research share by country group		
research_group	std_share	mean_share
United States	0.241679	0.451538
Germany	0.073499	0.098842
France	0.070118	0.075001
Other	0.067972	0.212052
United Kingdom	0.048863	0.129999
Netherlands	0.035698	0.038489

This table is to assess the stability of each country group's scientific prominence over time.

The United States exhibits by far the highest volatility (std = 0.24), reflecting large historical swings in its share of Nobel-level research. The sharp rise after 1945 and decline after 2000 contribute to this pattern of long-term amplitude.

Germany, France, the United Kingdom, and the Netherlands all show relatively low volatility (std = 0.03–0.07), indicating more stable long-run trajectories.

Critical Evaluation

Model Assumption

1. Main Research Country assumption

This project assumes that a laureate's main research country at the time of receiving the Nobel Prize can be used to infer their scientific mobility. However, this does not always reflect real scientific trajectories. Many laureates worked in several countries before winning the prize. In some cases, the country recorded as their "main research location" during the award year might not fully represent the mobility path.

2. Linear comparability

We have compared Nobel shares across time. But the 1901–1920 scientific community was tiny and is Europe-centered. High European shares then don't necessarily indicate greater talent concentration but also that may reflect limited global participation.

Data Limitation and deviation

For both models, in the data preprocessing stage, several historical entities such as Nazi Germany and German Reich were merged into the single modern entity Germany. However, this merging is essentially a simplifying assumption and may introduce certain biases. For example, different historical periods of Germany had substantial political, economic, and scientific institutional differences, and treating them as a single entity may artificially inflate Germany's overall node weight in the network.

Also, Six Nobel Prize categories follow very different research and career patterns:

- Physics/Chemistry/Medicine: Large laboratory infrastructures, high equipment dependence, typically with much stronger international mobility.
- Literature/Peace: Language-bound or politically embedded fields, typically with much lower mobility.
- Economics: Established from 1969 which is different to the original five prizes

We treat all categories together in our analysis, but the mechanisms driving mobility differ substantially

Lastly, Wikidata is a crowdsourced platform, which means institutional affiliation records are often incomplete and cannot fully reflect the actual patterns of scientific mobility

Discussion

1. Mechanisms of US Attraction

From the results, we found that the United States becomes the dominant destination for Nobel Prize-winning researchers, especially after the 1940s. A clear core-periphery structure emerges, with the US at the core and most other countries acting as peripheral senders.

Several factors may help explain this development. One major influence is the large outflow of scientific knowledge from Europe during the 1930s and 1940s. Political instability, and World War II displaced many leading scientists, who moved to the United States in search of safety and better research conditions. This historical played a crucial role in strengthening American scientific institutions.

In addition, the United States made significant strategic investments in research after the war. The government dramatically increased research funding, primarily allocated through new or expanded agencies such as the National Science Foundation and the National Institutes of Health. This stable and generous stream of funding created an unprecedented resource base for scientific research at universities.

Taken together, these factors suggest that the United States' leading position did not happen by chance. Instead, it reflects both major historical events and purposeful policy decisions. The mixture of wartime arrivals of talent and long-term investment in research institutions created a strong and lasting brain gain, which continues to influence the global distribution of Nobel-level scientific work today.

2. Low Modularity

This relatively low modularity score suggests that the global mobility network does not form strong regional clusters. Instead, the network is dominated by a single central hub, with most flows directed toward the United States. Such a strongly centralised, core-periphery structure naturally suppresses the formation of multiple balanced communities, which explains why the modularity value remains low. When modularity is very low, the detected communities lose substantive meaning, they mainly reflect algorithmic partitioning rather than real structural clusters.

Complementary Insights from Two Models

The social network analysis and the temporal model offer two different but complementary ways of understanding Nobel laureate mobility. SNA shows the main migration routes, highlights which countries act as hubs or remain on the periphery, and reveals community patterns. The temporal model instead focuses on how these patterns have changed over time, identifying major historical phases and long-term shifts in scientific leadership.

Comparison with Published Research

When comparing the mobility patterns observed in this study with the large-scale survey by Franzoni, Scellato and Stephan (2012) covering more than 17,000 scientists across 16 countries, we find remarkably consistent international mobility structures. In both datasets—despite their completely different populations—Switzerland exhibits the highest share of foreign-born scientists (GlobSci: 56.7%; this study: 58%), while the United States displays strong global attractiveness and a high reliance on foreign talent (GlobSci: 38.4%; this study: 39.8%).

Such cross-sample alignment indicates that the Nobel mobility patterns identified here not only characterize elite scientists but also mirror broader global scientific labour dynamics, thereby strengthening the recognisability and external validity of our findings.

Conclusion

This study examined the global geography of Nobel Prize-winning research from 1901 to 2024, analyzing 847 laureates across 75 countries using Social Network Analysis and temporal trend analysis.

Key Findings:

Our analysis reveals a profound transformation in scientific geography. The United States emerged as the overwhelmingly dominant hub, with net inflow exceeding 160 laureates. Traditional European centers (Poland, Germany, Austria) experienced substantial net outflows. This represents a shift from European dominance to US hegemony followed by a slight movement toward decentralisation after 2000.

Research Questions Answered:

RQ1: The US stands alone as the primary brain-gain nation. Brain-loss is more distributed across Poland, Germany, and Eastern Europe, with no single country experiencing loss comparable to US gain. The pattern is core-periphery rather than balanced exchange.

RQ2: Internationalization followed a U-shaped curve—high in early 20th century, declining mid-century during geopolitical tensions, rebounding to 35-40% post-1990. However, destinations fundamentally changed from multi-polar European circulation to US-dominated flows.

Reference:

Franzoni, C., Scellato, G., & Stephan, P. (2012). Foreign-born scientists: Mobility patterns for 16 countries. *Nature Biotechnology*, 30(12), 1250–1253.